

Teen brains

Throughout childhood, the brain grows and develops, and by the time a child is 6 years old, their brain has reached up to 90 percent of its full adult size. During puberty, however, the brain experiences its most dramatic transformation, in ways that can affect a teen's thoughts and behavior.

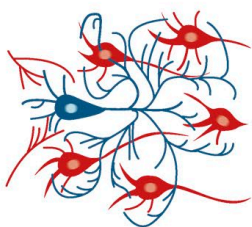
Changing brains

As children get older, their brains must learn how to cope with life as adults, and so transformations occur in the anatomy and chemistry of the brain. Areas involved in more basic functions, such as processing sensory input, mature first, while the areas responsible for more complex thought, such as planning ahead, develop later. These changes begin during puberty. Although the brain is fully grown by a person's mid-20s, it continues to develop and change for the rest of their life.

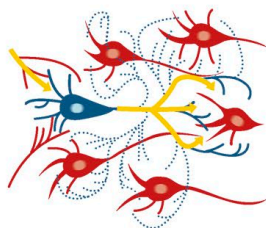
GOOD TO KNOW

Making connections

In early life, millions of connections called synapses form between brain cells, and these are what allow a person to learn. Over time, the brain starts to prune the connections that aren't used. This gives more space and "brain power" to those that are needed, making them more efficient. The teen years are a critical period for strengthening and pruning connections, but this process continues throughout a person's life.



◀ **Early childhood**
Connections form quickly, allowing a person to develop new skills.



◀ **Adolescence**
Unused connections fade away, while those that are used frequently become stronger.

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Prefrontal cortex
Responsible for rational thinking, problem solving, self-control, and thinking ahead, this is the last part of the brain to mature.

Basal ganglia
The basal ganglia controls movement and decision-making.

Nucleus accumbens
The brain's reward center, this helps form memories in response to positive or negative experiences. Dopamine levels in the nucleus accumbens change during adolescence—teens need more than adults to achieve the "buzz" from pleasure, meaning they'll take more chances to achieve this feeling.

Hypothalamus
This area triggers puberty by releasing the hormone GnRH. It also regulates sleep, body temperature, hunger, and thirst.

Pituitary gland
This controls hormone levels in the body.

Amygdala
Responsible for instinctive behavior, aggression, and risk-taking, the amygdala creates powerful emotions, such as fear and anger. Teens use the amygdala to process information more than adults, making them more prone to extreme emotions.

